From: Brady, Donal (EGLE)
Sent: 9/30/2022 9:32:14 AM
To: "Patrick Miller" <pmiller@cadillac-mi.net>; "Cindy Tomaszewski" <ctomaszewski@Cadillac-MI.net>
Cc: "'utilities@cadillac-mi.net'" <utilities@cadillac-mi.net>
Subject: Cadillac WWTP - NPDES Permit MI0020257 - September 27, 2022 - Compliance Communication CC004335
Attachments: Cadillac WWTP MI 0020257 CEI Report 092722.pdf

Hi Patrick and Cindy,

This email is the formal transmittal for the inspection we completed earlier this week. It was great meeting you Patrick. Thanks for completing the inspection in only your second week on the job.

On September 27, 2022, staff of the Michigan Department of Environment, Great Lakes, and Energy (EGLE), Water Resources Division (WRD), conducted an evaluation of Cadillac wastewater treatment plant (WWTP), located at 1121 Plett Road in Cadillac, Wexford County, to evaluate the facility's compliance with Part 31, Water Resources Protection, of the Natural Resources and Environmental Protection Act (1994 PA 451, as amended) and National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257. The facility (with an average day design flow of 3.2 MGD) discharges municipal wastewater, through Outfall 001, to the Clam River.

Patrick Miller and Cindy Tomaszewski from the City of Cadillac and Don Brady from EGLE, WRD participated in the inspection, which included an interview, records review, laboratory review, and a site walkthrough.

The following items were identified during the inspection and/or records review.

- The WWTP needs to install newly calibrated NIST thermometers in the sample refrigerators and automatic samplers and replace other expired NIST thermometers, as necessary
- The WWTP needs to evaluate the TSS procedure and incorporate reweighing of dried samples into the procedure, as necessary or required by the method
- The drying oven was set to a temperature of 110 degrees F. This slightly higher than normal temperature is necessary to dry the samples within a reasonable timeframe. The WWTP has verified the temperature does not result in sample loss. The WWTP should evaluate compliance with the method and consider different drying dishes and lower oven temperatures, if required by the method.
- The WWTP paused inter-lab split samples during the Covid pandemic. The WWTP should consider reinitiating the inter-lab split samples.
- Some foam was present at the discharge during the inspection. WRD understands the foam was unusual. The WWTP standard procedures for investigating the appearance of foam involves inspecting the upstream WWTP processes and collection system. In the past, foam appearance at the discharge has been transient. The WWTP shall monitor the discharge for the presence/absence of foam. Foam in the receiving stream may be a violation of the narrative water quality standard. Persistent or widespread foam in the receiving stream from the discharge shall be reported to EGLE, WRD in accordance with the WWTP’s NPDES permit.

- WRD understands the WWTP will be switching to the Hach TNT methods for phosphorus and ammonia. Please note, EGLE WRD approval is not needed for this change. However, the WWTP needs to evaluate the need for sample distillation when using the Hach TNT method for ammonia.

EGLE, WRD requests the following additional information. Please provide this information by October 31, 2022.

- The daily flow recorded on the August 31 bench sheet appeared slightly different than the flow reported in the DMR. Please check this flow and notify me if any corrective action is necessary.
- Please confirm the mercury, pH, and DO samples are collected after disinfection per the NPDES permit.
- Please let me know if the facility has a pollution incident prevention plan or other response plan that is intended to satisfy the requirements of Michigan's Part 5 regulation.

Other items…

- I strongly recommend the City of Cadillac submit a CWSRF form for any WWTP or collection system improvements that the City is planning. The CWSRF program is currently distributing grant dollars, including dollars for rural communities. The form is not a commitment but is a requirement in order for the City to qualify for CWSRF grants. The form is due November 1.
- I will send a follow up email regarding asset management. I know the City has implemented an asset management program and the City has provided me with much of the asset management information required by the NPDES permit. I simply need to see more detail and talk through some items with the City. Some items we should discuss include the City's capital improvement plan, headworks improvements, and alternatives to co-settling.

The WWTP and WWTP laboratory are well maintained and other items evaluated during the inspection were acceptable and in compliance with the permit. Thank you for your time during the inspection. If you have any questions or want to talk about this email, please let me know.

EGLE | MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

Donal Brady, P.E.
Environmental Engineer
Water Resources Division / Cadillac District Office
Michigan Department of Environmental, Great Lakes, and Energy
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ATTACHMENT NAME:

Cadillac WWTP MI 0020257 CEI Report 092722.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

Cadillac WWTP : 15637

NPDES - Facility Site Review

Inspector: Donal Brady
Start Date: 09/27/2022

Inspection Information

1. Permit Number: MI0020257

2. Date: 09/27/2022

3. Time: 11:00

4. Inspection Participants Present

Name	Title	Affiliation	Phone Number
Patrick Miller	Water Resources Supervisor	City of Cadillac	231-306-8084
Cindy Tomaszewski	Laboratory Supervisor	City of Cadillac	231-779-8089
Don Brady	Environmental Engineer	EGLE	231-429-5686

5. Opening Conference Notes:

Patrick introduced himself as the new WWTP supervisor (replacing Doug L.) and we discussed WWTP operations in general, needs, and future projects. We discussed the inspection procedure in general.

6. Select the current facility classification(s). See Help instructions if there is no facility classification on record.

Class A, treatment facilities serving or designed to serve a population in excess of 50,000 persons.

WWTP serves a population between 10,000 and 15,000 people but was upgraded to Class A due to complexity and discharge to a trout stream.

7. Designated Certified Operator

Certification Type	Operator Name	Number	Exp. Date	Reported to DEQ?
A, B, C, D	Jeff Dietlin	10435	04/15/23	X

Patrick is also a certified operator with B, C, D, and L1 licenses, #19313, expiration 01/15/25

8. If the facility is a mobile home park, do any residents own their own lot? ☐ Yes ☐ No ☑ NA

9. If applicable, population served by treatment plant (systems that serve the public only) 11,377

Per 2022 NPDES renewal application

10. Has the service area of wastewater plant changed since last application submittal? (public system only) ☐ Yes ☑ No ☐ NA

11. Identify chemicals used in treatment system

Chemical Name	Use/Purpose	Concentration	Location Feed Point	Authorized	None used
Ferric Chloride	Phosphorus removal	38.5%	Aeration / activated sludge basins	X	
Sodium hypochlorite	Periodic treatment of filamentous	90%	Various	X	

The WWTP may evaluate moving the FeCl feed to the downstream end of the aeration / activated sludge basins

Cadillac WWTP : 15637
NPDES - Facility Site Review
Inspector: Donal Brady
Start Date: 09/27/2022

Inspection Information

12. If the facility is a POTW, does it have an approved Industrial Pretreatment Program (IPP)? ☑ Yes ☐ No ☐ NA

13. If the POTW accepts septage, landfill leachate or other high strength waste, via truck or sewer, and does not have an IPP, how does it determine it can adequately treat this waste?
WWTP has IPP and is currently in the process of reevaluating the headworks loading.

14. Number of wastewater personnel recommended in the O&M Manual (public systems) 1

The O&M manual discusses shifts not # of personnel.

15. If the facility accepts septage, is it fully treated in the POTW? ☐ Yes ☐ No ☑ NA

The facility does not accept septage. The facility has 17 industrial users per IPP. The facility also accepts floor wash water from High Point Auto and also receives water drained from vactor truck waste (from sewer cleaning).

16. Does the facility have compliance issues that may be related to the acceptance of septage? ☐ Yes ☐ No ☑ NA

17. Does the facility have any plans to change any of their operations as it relates to the acceptance of septage? ☐ Yes ☑ No ☐ NA

18. What is the facility's onsite solids storage capacity design in months? Estimated to be approximately 1 year

19. What is the estimated remaining space currently in the storage structure? Estimated to be approximately 6 months

20. Solids are disposed of through? (choose all that apply): Land application

21. If the facility normally land applies, is there a back-up disposal plan in case they are unable to land apply? The backup plan would likely be disposal in a landfill (currently grit is disposed in a landfill)

22. Have they had any operation or compliance issues that were related to biosolids storage issues? ☐ Yes ☑ No ☐ NA

Specific Processes Producing Wastewater (Industrial/Commercial Only) & Treatment Unit Processes

23. If an industrial/commercial facility, identify the types of wastewater generated.

Municipal WWTP

Specific Processes Producing Wastewater (Industrial/Commercial Only) & Treatment Unit Processes

24. If an industrial/commercial facility, are these wastewater sources identified in the current permit application? ☐ Yes ☐ No ☑ NA

25. List the specific industrial/commercial processes that produce the raw wastewater flows:

Process Name
Municipal WWTP

General Observations/Questions

26. No. of wastewater treatment personnel 7

27. Does the number of wastewater treatment personnel appear to be adequate? ☑ Yes ☐ No

28. Number of hours each shift and/or day that the waste treatment system is staffed.

Shift #	Sun	Mon	Tues	Wed	Thur	Fri	Sat
Shift 1	Few hours	7:00 AM to 3:30 PM	7:00 AM to 3:30 PM	7:00 AM to 3:30 PM	7:00 AM to 3:30 PM	7:00 AM to 3:30 PM	Few hours
Shift 2							
Shift 3							

29. How are alarms/upsets monitored and responded to during off hours and on weekends? telephone auto-dialer

Dialer calls City offices during business hours and Central Dispatch during off hours, WWTP personnel are subsequently notified by City or Central Dispatch

30. List all treatment unit processes, including solids handling activated sludge, clarification, disinfection, equalization basin, aeration, attached growth, other (note other treatment below in comments), Sludge storage, Sludge stablization

Attached growth = RBCs used for tertiary treatment, sludge stabilization = anaerobic digestion, disinfection = UV, other = cloth filtration, downstream of RBCs, for tertiary treatment

31. Are all treatment units in service? If No, explain in comments below, including why, how long out of service, and when expected to be back in service. ☐ Yes ☑ No ☐ NA

One channel monster is currently out of service due to a worn shaft

General Observations/Questions

32. Any SSOs of the collection system or bypasses at the WWTP since the last inspection? ☐ Yes ☑ No ☐ NA

33. If yes, what is the date of the SSO/bypass and the date notification was made to EGLE?

Date of bypass	Date notification made to DEQ

34. If the facility is a POTW, are effluent flows at or above 80% of the design capacity? ☐ Yes ☑ No ☐ NA

35. Are hydraulic and/or organic overloads experienced? If Yes, explain in comments. ☑ Yes ☐ No ☐ NA

A transient organic overload was experienced recently

36. Has there been an increase in flow since your last permit was issued? ☐ Yes ☑ No

37. If an industrial or privately owned facility, is the current effluent flow rate greater than that authorized by the current permit? ☐ Yes ☐ No ☑ NA

38. Is a permit modification needed? ☐ Yes ☑ No

39. Identify the influent and effluent flow monitoring points.

Monitoring Point	Location	Acceptable
Effluent	After secondary clarifiers	X

40. Influent flow is measured before all return lines? (public system only) If No, where is it measured? ☐ Yes ☐ No ☑ NA

41. The effluent flow measurement represents all wastewater being discharged (e.g. after all return lines)? ☑ Yes ☐ No

42. Select type of effluent flow metering weir

43. When was it last calibrated? 08/15/2022

General Observations/Questions Cont.

44. Has the facility received any citizen complaints or experienced any spills since the last inspection? ☐ Yes ☑ No ☐ NA

45. Any unusual conditions noted in treatment system effluent, receiving water, irrigation fields, infiltration basins, or tile fields? If Yes, describe in comments section. ☐ Yes ☑ No ☐ NA

46. Any evidence of bypasses or chemical spills, surcharges and/or overflow in the wastewater collection or treatment system? If Yes, describe in comments section. ☐ Yes ☑ No ☐ NA

Normal bypasses of portions of the collection system occur during construction projects (e.g. sewer replacements)

General Observations/Questions Cont.

	Yes	No	NA
47. Any accumulation of solids, grease, foam or floating material in wetwells, sumps, clarifiers, land application areas, or other treatment units?	☐	☑	☐
48. Any excessive discoloration, puddles, oil or grease on ground or floor near equipment or tanks?	☐	☑	☐

49. Identify in the table below all significant chemical storage areas.

Chemical Name	Location	Concentration	Volume
Ferric Chloride	Room/building next to UV	38.5%	Two tanks each approximately 8,200 gallons

	Yes	No	NA
50. Does the Facility conduct observations of their chemical use and storage areas to determine if leaks or discharges have occurred?	☑	☐	☐
51. Does secondary containment for outdoor storage of chemicals appear to have sufficient capacity, be compatible with the material stored in it, and be capable of containing the material should a spill occur?	☐	☐	☑
52. Does indoor storage of chemicals appear to have sufficient curbing, containment, or is situated such that a spill would not reach public sewer systems or surface / groundwaters?	☑	☐	☐
53. Does the facility have a Pollution Incident Prevention Plan (PIPP) or Integrated Contingency Plan (ICP)?	☑	☐	

Facility will confirm this

	Yes	No	NA
54. Any problems with wastewater treatment chemical feed system? If Yes, describe in comments section.	☐	☑	☐
55. Has the facility reported chemical spills that have occurred?	☐	☐	☑
56. Any jury rigged or temporary systems to correct operation problems? If Yes, describe in comments	☐	☑	☐
57. Any obnoxious odors noted? If Yes, describe from where in comments.	☐	☑	☐
58. Were any unidentified/unmarked confined spaces observed? If Yes, describe where in comments.	☐	☑	☐
59. Is lighting adequate for safe operation and observation?	☑	☐	
60. Are aisles, ladders, stairs and other access points kept clear and in good condition?	☑	☐	☐
61. Does general housekeeping appear to be adequate?	☑	☐	
62. Does the treatment system security/fencing appear to be adequate?	☑	☐	☐
63. Were photographs taken?	☑	☐	

General Observations/Questions Cont.

64. Have all deficiencies previously cited in inspections, enforcement actions, correspondence and verbal communications been corrected? ☐ Yes ☑ No ☐ NA

The WWTP is working to address IPP items and provide additional asset management information.

65. Is a permit modification in order due to a change in name, ownership, or operations? ☐ Yes ☑ No ☐ NA

Closing Information

66. Completed By: Don Brady

67. Closing Conference Notes:

The WWTP is well maintained and the City is doing a good job with O&M, monitoring, and reporting
Some items to address include the following
- Install NIST thermometers in WWTP sample refrigerators and samplers, replace expired NIST thermometers
- Evaluate reweighing dried samples during solids analysis
- Evaluate different drying dishes and drying oven temp
- Restart inter-lab split sample analysis
- Send corrective action information for DMR/QA 39
- Evaluate foam at discharge
- Evaluate alternatives to co-settling
- Evaluate need for spill response plan and Part 5 compliance
- Submit a CWSRF ITA form by November 1
- Submit and discuss additional asset management information to WRD
- WRD will check on approval needed (if any) for switching to Hach P and NH4 analysis, WRD will also send more information on Part 5 and asset management

Desk review after inspection indicated the following
- 8/31/22 bench sheet flow vs. DMR needs to be checked
- WWTP should confirm mercury, pH, and DO sample locations are per permit
- Additional information on the effluent weir should be discussed during future inspections
- WRDs Municipal/Public Operation and Maintenance Inspection forms should be discussed during future inspections

Cadillac WWTP : 15637
NPDES - Self-Monitoring Records
Inspector: Donal Brady
Start Date: 09/27/2022

General Information

1. Facility Designated Name:	Cadillac WWTP
2. Permit Number:	MI0020257
3. Date:	09/27/2022
4. Time:	11:00 AM

Record Information

5. Are all records and information resulting from the monitoring activities required by this permit maintained for at least three years? Examples include DMR's, MORs, lab data, chain of custody forms, calibration & maintenance records and recordings from continuous monitoring instrumentation.

☑ Yes ☐ No ☐ NA

Use the table(s) in the following section to spot check at least three sets of data points (including geometric means, loadings, averages, etc.).

Analytical Tables

6. Use the table below to spot check sets of data points (including geometric means, loadings, averages, etc.).

Date	Sampling Point:	Type of Wastewater	Parameter	Reported Result (from DMR)	Site Record (from bench sheet or lab)	Accurate?
08/31/2022	After Tertiary	Municipal	P	0.20 mg/L	0.199 mg/L	X
08/31/2022	After Tertiary	Municipal	TSS	4.00 mg/L	4 mg/L	X
08/31/2022	After Tertiary	Municipal	NH4	<0.300 mg/L	ND	X
08/31/2022	After Secondary Clarifiers	Municipal	Flow	2.02 MGD	2.048 MGD	
08/31/2022	Outfall	Municipal	DO	7.8 mg/L	7.81 mg/L	X

Flow appears slightly off, ask WWTP to check

7. Are analytical results/bench sheets consistent with the data reported on the DMRs? Explain any discrepancies in the comments.

☑ Yes ☐ No ☐ NA

Record Information Cont.

8. Does the facility have authorization for elimination of, or reduced monitoring? If yes, identify in the Comment Section which parameters have been reduced or eliminated along with their "new" monitoring

☐ Yes ☑ No

9. Monitoring Records are adequate and include:

Parameter	Check if included
Dates, times, frequency and location of sampling collection	X
Name of individual(s) performing sampling	X
Results of analyses	X
Dates of analyses	X
Name of person(s) performing analyses	X
Analytical technique or method used	X
The date of and name of person responsible for equipment calibration	X

Cadillac WWTP : 15637

NPDES - Self-Monitoring Records

Inspector: Donal Brady

Start Date: 09/27/2022

Record Information Cont.

Question	Yes	No	NA
10. Are laboratory bench sheets and other records understandable and clearly filled out?	☑ Yes	☐ No	☐ NA
11. Does the inspection contact/operator have a current copy of the permit?	☑ Yes	☐ No	☐ NA
12. Can the permittee produce a copy of their WTA approvals along with documentation that the factors associated with the approval have not changed (e.g. dosage rates, flow volumes, bench test calculations, etc.)?	☐ Yes	☐ No	☑ NA

WTA approved per Part 41

Closing Information

13. Closing Conference Notes:

See facility site review form

14. Completed By:	Don Brady

Cadillac WWTP : 15637

NPDES - Sampling Procedures

Inspector: Donal Brady
Start Date: 09/27/2022

General Information

1. Facility Designated Name:	Cadillac WWTP
2. Permit Number:	MI0020257
3. Date:	09/27/2022
4. Time:	11:00 AM

5. Sampling Parameters (if answer is not available, use comments section):

Parameter	Sampling Location	Sampling Equipment (auto-sampler, etc)	Sample Type (grab, composite, etc)	Sampling Location Representative?	Sampling Container Material and Quality	Preservation Method	Holding Time
NH3-N	After tertiary	Autosampler	Composite	X	HDPE	Cool	28 days
TSS	After tertiary	Autosampler	Composite	X	HDPE	Cool	7 days
(C) BOD5	After tertiary	Autosampler	Composite	X	HDPE	Cool	48 hours
Total Phosphorus	After tertiary	Autosampler	Composite	X	HDPE	H2SO4, cool	28 days
Fecal Coliform	Outfall		Grab	X	Polystyrene	Cool	8 hours
Hg	Outfall		Grab	X	Lab supplied	Cool	72 hours
pH	Outfall		Grab	X			15 mins
DO	Outfall		Grab	X			15 mins
Metals	After tertiary		Composite	X	Lab supplied	NO3, Cool	6 months
WET	Completed for cycle						
Bis-2-ethylhexyl-phthalate	After tertiary		Composite	X	Lab supplied	Cool	7 days
PFOS/PFOA	After tertiary		Grab	X	Lab supplied	Cool	14 days

Ask WWTP to confirm Hg, pH, and DO collected after UV

Inspection Information

6. How are staff trained to collect samples? — One-on-one w/ experienced staff, On-the-job training

7. Are sample collection procedures reviewed periodically? — ☑ Yes ☐ No

8. Are sampling procedure changes made when appropriate? — ☑ Yes ☐ No

9. Are 24-hour composite samples flow-proportional? If No, explain how it is representative of the discharge in the comments. — ☑ Yes ☐ No ☐ NA

10. Has an alternate sample type been approved?

Parameter	Sample Type

NA

11. Does the facility have an approved alternate monitoring location? — ☐ Yes ☑ No

Composite Sampler Information

12. Monitoring Point Location: — NA

Cadillac WWTP : 15637
NPDES - Sampling Procedures

Inspector: Donal Brady
Start Date: 09/27/2022

Composite Sampler Information

13. Review and/or inspect the composite sampler(s).

Question	Yes	No	NA
Is the temperature in the composite sampler 0-6°C?	X		
Does the sample line maintain a minimum velocity of 2ft/s?	X		
Is a line purging cycle used between samples?	X		
Any obstructions in the sample line?		X	
Does the length or number of bends in the sample line interefere with obtaining a representative sample?		X	

Composite samplers set to less than 6 degrees C, thermometers in samplers need to be replaced

Inspection Information Cont.

14. Is the laboratory refrigerator, or any equipment used to store samples, kept at a temperature between 0-6°C? If not, explain in the comments. ☑ Yes ☐ No

15. Does the facility conduct sampling in accordance with the NPDES permit language or 40 CFR Part 136? ☑ Yes ☐ No

16. Is monitoring and analyses performed more often than required by the permit? ☑ Yes ☐ No ☐ NA

Yes, for process control

17. If Yes, are the results included in the facility's monthly reports? ☐ Yes ☑ No ☐ NA

Effluent sampled per permit, other samples collected in plant for process control

Closing Information

19. Closing Conference Notes:
See facility site review form

20. Completed By: Don Brady

Cadillac WWTP : 15637
NPDES - Analytical Methodology
Inspector: Donal Brady
Start Date: 09/27/2022

General Information

1. Date:	09/27/2022
2. Time:	11:00 AM
3. Facility Designated Name:	Cadillac WWTP
4. Permit Number:	MI0020257

Laboratories

5. Identify all lab(s), including the in-house and contract labs, that perform the monitoring required by the NPDES Permit and list the parameters analyzed by that lab in the table below.

Name of Lab	Parameter(s)
Pace Analytical	Hg, Bis 2-ethylhexyl phthalate, conventional biosolids parameters
Eurofins	PFAS
Fibertec	PFAS
ERM	WET
HML	Biosolids pathogens

6. Evaluate the chain of custody record required between the permittee and a contract laboratory for the following elements.	Facility name, Sample date, Sample location, Sampler name, Number of samples, Parameters to be analyzed, Preservative used, Intake temperature, Signatures, Dates and times
7. Does the contract lab have a QA/QC Program?	☑ Yes ☐ No ☐ NA
8. Does the permittee periodically review the QA/QC program associated with the analyses performed by the contract lab to ensure it is adequate?	☑ Yes ☐ No ☐ NA

Cadillac WWTP : 15637

NPDES - Analytical Methodology

Inspector: Donal Brady
Start Date: 09/27/2022

Analytical Methods Used

9. Analytical Methods Table

Parameter	Method Used	Approved/Acceptable Method	Comments
CBOD	SM 5210B	X	
TSS	SM 2540 D	X	
Ammonia	SM 4500-NH3 D	X	
Phosphorus	SM 4500-P E	X	
Fecal Coliform	Colilert-18	X	
PFOS/PFOA	EPA 0537.1 (Modified)	X	Chemicals not listed in 40 CFR 136
Cu, Ni, Se	EPA 200.8	X	
Bis 2-ethylhexyl phthalate	EPA 625.1	X	
Hg	EPA 1631E / EPA 245.1	X	
pH	SM 4500 H+ B	X	
DO	SM 4500 O G	X	

10. Select one or more parameters to evaluate the protocols and/or techniques used during the analysis to determine if it meets the requirements of the test method.

Parameter	Testing protocols acceptable?
Fecal Coliform Colilert-18	X

WWTP is using IDEX 18 hour method for fecal coliform.

11. Approval has been obtained for alternative test methods? ☐ Yes ☐ No ☑ NA

12. Facility has reference manuals for the test methods that they are using? ☑ Yes ☐ No ☐ NA

13. Facility has written procedures or manuals for instruments and equipment use. ☑ Yes ☐ No ☐ NA

14. Are laboratory calibration records maintained? ☑ Yes ☐ No ☐ NA

Instrumentation

*15. List the analytical instruments the facility uses in their self-monitoring activities. *Confirm acceptability of calibration method using manufacturer specifications or test method requirements*

Instrument	Calibration Method*	Frequency	Date last serviced
LBOD probe (luminous) (CBOD)	Operator calibration curve per method	Daily	Within last 1 year
Balance	Third party contractor	Annually	Within last 1 year
ICP mass spectrometer (metals)	Operator calibration curve per method	Daily	Within last 6 months (serviced 2X/year in house)
Spectrophotometer (P and NH4)			Brand new instrument
pH/DO meter	Operator calibration curve per method	Weekly	Within last 1 year

16. Do laboratory instruments and equipment appear to be in good condition? ☑ Yes ☐ No ☐ NA

Cadillac WWTP : 15637
NPDES - Analytical Methodology
Inspector: Donal Brady
Start Date: 09/27/2022

Instrumentation

17. Are records kept with dates of laboratory equipment maintenance/repair/replacement? ☑ Yes ☐ No ☐ NA

18. Proper volumetric glassware is used? ☑ Yes ☐ No ☐ NA

19. Glassware is properly cleaned and sterilized, if necessary? ☑ Yes ☐ No ☐ NA

20. Select the type of analytical balance used. If the balance used it "Other", describe it in the comment section. Electronic

21. Is the balance clean and free of loose powders, stains, and corrosion? ☑ Yes ☐ No ☐ NA

22. If the balance is not equipped with automatic vibration compensation, is it mounted on a solid, vibration free surface, isolated from excessive heat, cold, or drafts? ☑ Yes ☐ No ☐ NA

23. Equipment temperature log maintained for ovens, incubators, etc. ☑ Yes ☐ No ☐ NA

24. Equipment temperatures. Indicate whether each is acceptable by checking Yes, No, or NA.

Equipment and allowable temperature	Actual temperature	Yes	No	NA
BOD incubator at 20°C? (±1°)	19.9 to 20.1	X		
Fecal coliform incubator at 44.5°C? (±0.2°)	44.5 (bath for IDEX trays)	X		
Drying oven for SS at 103-105°C? (±1°)	110	X		

The WWTP turned up to the drying oven as the samples were taking a long time to dry at lower temperatures, the WWTP verified the higher temperature did not result in sample mass loss.

25. Has the dessicant used with the suspended solids analysis reached its moisture saturation point? ☐ Yes ☑ No ☐ NA

26. Are temperatures verified with a National Institute of Standards and Technology (NIST) thermometer or NIST verified thermometer? ☑ Yes ☐ No ☐ NA

NIST thermometers are used. The thermometers in the sample refrigerators and samplers expired, the WWTP will replace expired thermometers.

Quality Assurance/Quality Control (QA/QC)

27. Are QA/QC written procedures in place for all required monitoring? ☑ Yes ☐ No ☐ NA

28. Are QA/QC results _______________ on control charts? recorded

Control charts/tables maintained for CBOD, P, and NH4

29. If data is plotted on control charts, does the data stay within upper and lower control limits? ☑ Yes ☐ No ☐ NA

Cadillac WWTP : 15637
NPDES - Analytical Methodology
Inspector: Donal Brady
Start Date: 09/27/2022

Quality Assurance/Quality Control (QA/QC)

30. If QA/QC data are outside of control limits, what was done to identify the cause?

Replace reagents, Check equipment, Create a new phosphorus curve, Clean glassware, Check laboratory water, Review testing protocols, Review data for accuracy, Other (provide information in comments section)

Other - conduct analysis with a reference sample

31. Were the QA/QC issues adequately addressed? ☑ Yes ☐ No ☐ NA

32. Using the same parameters previously reviewed under the Analytical Methods section, what QA/QC methods are used? If not applicable, put NA in the first parameter box and delete all other pre-populated parameters.

Parameter	Duplicates Frequency	Spikes Frequency	Blank/GGA Frequency	Slope Check	Reference Samples Frequency	Inter-lab Splits Frequency
CBOD (D,GGA,R,I)	1X / week		1X / day		1X / year	Periodically
TSS (D,R,I)	1X / week		1X / week		1X / year	Periodically
Ammonia as N (D,S,Slope,R,I)	1X / week	1X / week	1X / week	1X / day	1X / year	Periodically
Total phosphorus (D,S,R,I)	1X / week	1X / week	1X / week	1X / week & Cal curve / 6 wks	1X / year	Periodically
pH (D,Slope,R)	1X / week			Cal curve 1X / week	1X / year	
Fecal coliform (D,B)	1X / week	1X / week (+ check)	1X / week		1X / year	

WWTP suspended inter-lab splits during Covid pandemic but will re-start this analysis.

33a. If an EPA major or selected minor, has the facility participated in the EPA DMR/QA program? ☑ Yes ☐ No ☐ NA

33b. If yes, characterize their result from the list below. Use the Comments to explain results, if needed.

Acceptable, Not Acceptable - Adequate Corrective Action

WRD district staff need to upload the DMR/QA information into MiWaters. WRD asked for the corrective action documentation for DMR/QA 39, 40, and 41 be provided. The WWTP provided additional information for DMR/QA 40 and 41 during the inspection and will send additional information for DMR/QA 39.

34. Are there chemicals in the laboratory with an expired shelf life? If the chemical is associated with a parameter being monitored for the NPDES permit, identify the name of the chemical and the expiration date of that chemical in the Comment Section. ☐ Yes ☑ No ☐ NA

Cadillac WWTP : 15637

NPDES - Analytical Methodology

Inspector: Donal Brady

Start Date: 09/27/2022

Quality Assurance/Quality Control (QA/QC)

Laboratory Data Handling and Personnel

35. Significant figures are established for each analysis and round-off rules are uniformly applied?	☑ Yes ☐ No ☐ NA
36. Does staffing appear to be sufficient? If No, provide an explanation in the Comment Section.	☑ Yes ☐ No ☐ NA
37. Is there sufficient time allowed for analyses, QA/QC, laboratory maintenance, and recordkeeping? If No, provide an explanation in the Comment Section.	☑ Yes ☐ No ☐ NA
38. How are lab analysts trained?"	On-the-job training
39. Any changes in laboratory personnel since the last inspection? If Yes, identify what the change was and how (if at all) it has affected laboratory operations in the Comment Section.	☐ Yes ☑ No ☐ NA

Closing Information

40. Closing Conference Notes: See facility site review form	
41. Completed By:	Don Brady

Cadillac WWTP : 15637

NPDES - Flow Measurement

Inspector: Donal Brady
Start Date: 09/27/2022

Inspection Information

1. Facility Designated Name:	Cadillac WWTP
2. Permit Number:	MI0020257
3. Date:	09/27/2022
4. Time:	11:00 AM
5. Does the facility have loading limits authorized in their permit?	☑ Yes ☐ No
If No, complete the Type I Outfall section. Definition: These outfalls have no loading limits (e.g. pounds per day, pounds per month) in the NPDES permit.	
If Yes, complete the Type II Outfall or Type III Outfall section.Definition: Type II Outfalls have in-line flow meters and loading limits. Definition: Type III Outfalls have weir and/or flumes and loading limits.	

Type I Outfall

1. What flow measurement device(s) does the facility utilize? Estimation of flows is acceptable through pump curves, water meters, mass balance, use summaries, or similar techniques.

Outfall Number	Flow Measurement Method	Data Recording Device

2. Does the facility use pump curves or pump ratings?	☐ Yes ☐ No ☐ NA
2a. How old is the pump?	
2b. How are pumping times obtained?	
2c. When was the pump curve last calibrated/checked?	
2d. Who calculated the pump curve?	
2e. Have there been any maintenance problems with the pump? If yes, describe in the comments.	☐ Yes ☐ No ☐ NA
2f. Does the pumping rate vary significantly during discharge periods?	☐ Yes ☐ No ☐ NA
2g. If yes, is a head-discharge curve used?	☐ Yes ☐ No ☐ NA
2h. Do the conversion calculations for changing pump times to flow make sense?	☐ Yes ☐ No ☐ NA
2i. Are pressure gauges installed, read and recorded?	☐ Yes ☐ No ☐ NA
2j. If pressure gauges are used, is the height from the water surface to the gauge added to the gauge reading for total head?	☐ Yes ☐ No ☐ NA
3. Are water meters used?	☐ Yes ☐ No ☐ NA

Inspector: Donal Brady
Start Date: 09/27/2022

Type I Outfall

3a. When was the meter installed?

3b. Is the meter ever calibrated against an alternate method of flow measurement at least annually? ☐ Yes ☐ No ☐ NA

3c. Is the meter installed such that it remains full of liquid? ☐ Yes ☐ No ☐ NA

4. Are usage summaries used? ☐ Yes ☐ No ☐ NA

4a. When was the summary compiled?

4b. Who developed the summary?

4c. Have changes occurred in plant operations since the summary was compiled? ☐ Yes ☐ No ☐ NA

4d. Has the summary ever been compared against an alternate method of flow measurement? ☐ Yes ☐ No ☐ NA

5. Is the appropriate flow measurement method used for all Type I Outfalls?   If not, describe which outfall and why it is not acceptable in the ☐ Yes ☐ No ☐ NA

Type II Outfall

1. What flow measurement device(s) does the facility utilize? It is acceptable to use magnetic meters, doppler meters, venturi meters, flow nozzle meters, and orifice meters.

Outfall Number	Flow Measurement Method	Data Recording Device

2. Does the facility use a magnetic meter(s)? Magnetic Meters - A voltage is created when the flow passes through an electromagnetic field.   The voltage is proportional to the velocity of the flow (Faraday's Law.) ☐ Yes ☐ No ☐ NA

2a. Is the flow velocity in the pipe at least three to five feet per second? ☐ Yes ☐ No ☐ NA

2b. Are there any changes in the conductivity of the wastewater during the day (variations in dissolved solids - particularly chlorides?) ☐ Yes ☐ No ☐ NA

3. Does the facility use a doppler meter(s)? Doppler Meters - When an ultrasonic wave is reflected from bubbles, suspended solids, and turbulence in a moving fluid, the frequency shift of the reflected wave will be in proportion to the velocity of the fluid. ☐ Yes ☐ No ☐ NA

3a. If yes, are there significant changes in the suspended solids contents in the wastewater? ☐ Yes ☐ No ☐ NA

4. Does the facility use a venturi, flow nozzle or orifice meter(s)? Venturi, Flow Nozzle and Orifice Meters - The flow rate is determined by the pressure differential across the meter. ☐ Yes ☐ No ☐ NA

Cadillac WWTP : 15637
NPDES - Flow Measurement
Inspector: Donal Brady
Start Date: 09/27/2022

Type III Outfall

1. What flow measurement device(s) does the facility utilize? It is acceptable to have either weirs or flumes. Liquid flows over weirs, but through flumes.

Outfall Number	Flow Measurement Method	Data Recording Device
001	Weir	Level transmitter

2. Does the facility use a weir(s) for flow measurement?	☑ Yes ☐ No ☐ NA
2a. What type of weir is used?	v-notch
2b. Is there excessive sediment build-up behind the weir plate?	☐ Yes ☑ No ☐ NA
2c. Is there leaking around or under the weir plate?	☐ Yes ☑ No ☐ NA
2d. Has debris (algae, sticks, paper) collected on the weir crest?	☐ Yes ☑ No ☐ NA
2e. Is the weir plate perpendicular to the direction of the flow?	☑ Yes ☐ No ☐ NA
2f. Does the weir plate appear level?	☑ Yes ☐ No ☐ NA
2g. Does the weir plate appear to be plumb?	☑ Yes ☐ No ☐ NA
2h. Does the water flow freely and clearly over the weir crest (can an air space be seen below the nappe of the water?)	☑ Yes ☐ No ☐ NA
2i. Is the upstream flow channel straight for at least four times the depth of the water level in the channel?	☐ Yes ☐ No ☑ NA
Unknown, WRD will evaluate further during future inspections	
2j. Is the distance from the weir opening to the side of the channel at least two times the head flowing over the weir?	☐ Yes ☐ No ☑ NA
Unknown, WRD will evaluate further during future inspections	
2k. Does the velocity of approach in the channel cause turbulence near the weir?	☐ Yes ☑ No ☐ NA
2l. How are head measurements taken?	Level transmitter
2m. If a bubbler tube or an ultrasonic level measuring device is used, how is the device calibrated?	Transmitter calibrated annually by third party contractor
2n. How often is it calibrated?	Annually
2o. What flow measurement formula is the permittee using?	Unknown, WRD will evaluate further during future inspections
3. Does the facility use a flume(s) for flow measurement?	☐ Yes ☑ No ☐ NA

Cadillac WWTP : 15637
NPDES - Flow Measurement

Inspector: Donal Brady
Start Date: 09/27/2022

Type III Outfall	
3a. Is the channel approach to the flume straight?	☐ Yes ☐ No ☐ NA
3b. Is the flow depth across the channel uniform?	☐ Yes ☐ No ☐ NA
3c. Are there any waves or turbulent conditions in the approach channel?	☐ Yes ☐ No ☐ NA
3d. Where is the level (head) measurement taken?	
3e. How is the level measured?	
3f. If a float device is used, is a stilling well used?	☐ Yes ☐ No ☐ NA
3g. If a bubbler tube, or ultrasonic device is used, how is the device calibrated?	
3h. How often is it calibrated?	
3i. Does the water flow freely through the throat of the flume?	☐ Yes ☐ No ☐ NA
3j. If the flow does not appear to be free, how does the permittee measure head?	
3k. What flow measurement formula is the permittee using?	

Closing Information	
Closing Comments: See facility site review form	
Completed By:	Don Brady
Date of Inspection:	09/27/2022

Cadillac WWTP : 15637
NPDES - Unit Processes Checklist
Inspector: Donal Brady
Start Date: 09/27/2022

Unit Processes Checklist

1. Facility Name:	Cadillac WWTP
2. Preliminary Treatment	Flow Equalization Basin, Comminutor/Barminutor, Mechanical Grit Chamber
3. Primary Clarifier	Circular
4. Treatment Types	Activated Sludge
5. Secondary Clarifiers	Circular
6. Disinfection	Ultra-Violet Radiation
7. Sludge Storage	Sludge Storage Tank
8. Sludge Thickening	
9. Sludge Dewatering	Drying Beds
Drying beds are functional but not used for biosolids.	
10. Sludge Stabilization	Anaerobic Digestion
11. Tertiary Treatment	
RBCs and cloth filters	
12. Tertiary Treatment (Specify Treatment Type)	RBCs and cloth filters

Cadillac WWTP : 15637

NPDES - Areas Evaluated

Inspector: Donal Brady
Start Date: 09/27/2022

Areas Evaluated	
1. Permit Review	Not Applicable
2. Compliance Schedule	Not Applicable
3. Facility Site Review	Satisfactory
4. Effluent/Receiving Waters	Marginal
Some foam observed at discharge, this is not normal according to WWTP personnel	
5. Records/Reporting (Off-site review)	Satisfactory
6. Self-Monitoring Program (On-site review)	Satisfactory
7. Sample Protocol	Satisfactory
8. Laboratory	Satisfactory
9. Biosolids/Sludge Disposal (Optional)	
10. Flow Measurement (Optional)	Satisfactory
11. Operation & Maintenance (Optional)	Satisfactory
12. Other	